PUNJAB ENGINEERING COLLEGE (DEEMED TO BE UNIVERSITY)
Sector 12, Chandigarh - 160012 | Office of Dean Academic Affairs (DAA)
Capstone Coursework Deliverable • Academic Clearance Automation Engine

Software Requirements Specification (SRS)

Academic Grievance, No-Dues & Clearance Automation Engine

Institution: Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh - 160012
Document Identifier: PEC-SE-SRS-FROZEN
Standard Compliance: IEEE Std 830-1998 (Recommended Practice for Software Requirements Specifications)
Baseline Version: 3.0.0 (FROZEN ARCHITECTURAL BASELINE)
Administrative Authority: Office of the Dean Academic Affairs (DAA), PEC Chandigarh
Classification: Institutional Standard Specification
Status: FROZEN & RATIFIED (Implementation-Agnostic Baseline)

Baseline Freeze Declaration & Change Control Policy

> **NOTICE OF BASELINE FREEZE:**
> This specification defines the immutable architectural, functional, data, and behavioral contracts for the PEC Academic Grievance, No-Dues & Clearance Automation Engine. All functional boundaries, mathematical formulas, state transitions, security constraints, and data definitions in this document are **FROZEN**.
>
> Any future implementation-level modifications—including UI component updates, styling alterations, backend database driver migrations (e.g., SQLite to PostgreSQL), or API transport optimizations—are explicitly encapsulated within the interface abstractions defined herein and **shall not invalidate or alter this SRS, the system DFDs, or the UML structural and behavioral models**.

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to establish the formal, complete, and verifiable requirements for the **Academic Grievance, No-Dues & Clearance Automation Engine** deployed at **Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh**.

This document serves as the contractual baseline between institutional stakeholders (Students, Department In-Charges, Dean Academic Affairs, and Finance Branch) and the software engineering team, governing the digital transformation of institutional clearance workflows.

1.2 Scope of the System

The system automates the multi-department no-dues clearance and grievance arbitration workflows across the Sector 12 campus:

1. **Identity & Positional Authority Resolution:** Authentication of PEC students using unique 8-digit Student Identification Numbers (SIDs, e.g., `22103042`) and department authorities via permanent administrative titles (*Dean Academic Affairs*, *Chief Warden Hostels*, *Librarian*, *President Gymkhana*, *HOD Computing & Labs*, *Accounts Officer*).
2. **Multi-Gate Approval Pipeline (Chain of Responsibility):**
   - **Gate 1 (Concurrent / Parallel):** Dr. B.R. Ambedkar Central Library, PEC Hostels (Shivalik, Kurukshetra, Himalaya, Aravali, Vindhya, Kalpana Chawla), and Gymkhana & Sports Complex.
   - **Gate 2 (Sequential / Prerequisite Barrier):** Departmental Academic and Research Laboratories (CSE, ECE, Mechanical, Electrical, Civil, Aerospace, Metallurgy, Production).
   - **Gate 3 (Terminal Financial Settlement):** Accounts & Finance Branch for semester fee reconciliation and Caution Money security deposit refund release.
3. **Cryptographic Validation & Digital Seals:** Keyed-Hash Message Authentication Code ($\mathrm{HMAC\text{-}SHA256}$) seals per departmental sign-off and SHA-256 integrity digests for issued clearance certificates.
4. **SLA-Driven Grievance Arbitration:** Formal dispute submission with priority tiers (Critical, High, Medium, Low) and automated escalation to the Dean Academic Affairs upon SLA deadline breach.
5. **Public Verification Gateway:** Unauthenticated public portal enabling employers, universities, and visa authorities to independently verify certificate authenticity.

1.3 Definitions, Acronyms, and Abbreviations

- **PEC:** Punjab Engineering College (Deemed to be University), Chandigarh.
- **SID:** Student Identification Number (8-digit unique institutional identifier, format: `YYBBBRRR`, e.g., `22103042`).
- **DAA:** Dean Academic Affairs, Punjab Engineering College.
- **DSA:** Dean Student Affairs, Punjab Engineering College.
- **Central Library:** Dr. B.R. Ambedkar Central Library, PEC Sector 12 campus.
- **Hostel Council:** Administrative body governing PEC student residences (Shivalik, Kurukshetra, Himalaya, Aravali, Vindhya, KCH).
- **Gymkhana:** PEC Sports Complex, Gymnasium, Grounds, and Swimming Pool facility.
- **Caution Money:** Refundable institutional security deposit collected at admission and reimbursed at Gate 3.
- **CoR:** Chain of Responsibility Behavioral Design Pattern (GoF).
- **SLA:** Service Level Agreement governing maximum turnaround times for student grievance disputes.
- **HMAC:** Keyed-Hash Message Authentication Code (RFC 2104 / FIPS PUB 198-1).
- **INR / ₹:** Indian Rupee, the official monetary unit of the institution.

1.4 References

- IEEE Std 830-1998: *IEEE Recommended Practice for Software Requirements Specifications*.
- FIPS PUB 180-4: *Secure Hash Standard (SHS)*, National Institute of Standards and Technology.
- RFC 2104: *HMAC: Keyed-Hashing for Message Authentication*.
- RFC 7519: *JSON Web Token (JWT)*.
- PEC Academic Ordinances & Regulations for B.Tech., M.Tech., and Ph.D. Degree Programs.

- UGC / AICTE Guidelines on Transparent Digital Grievance Redressal in Higher Educational Institutions.

1.5 Overview of the Document

The remainder of this document is organized according to the IEEE Std 830-1998 standard structure:

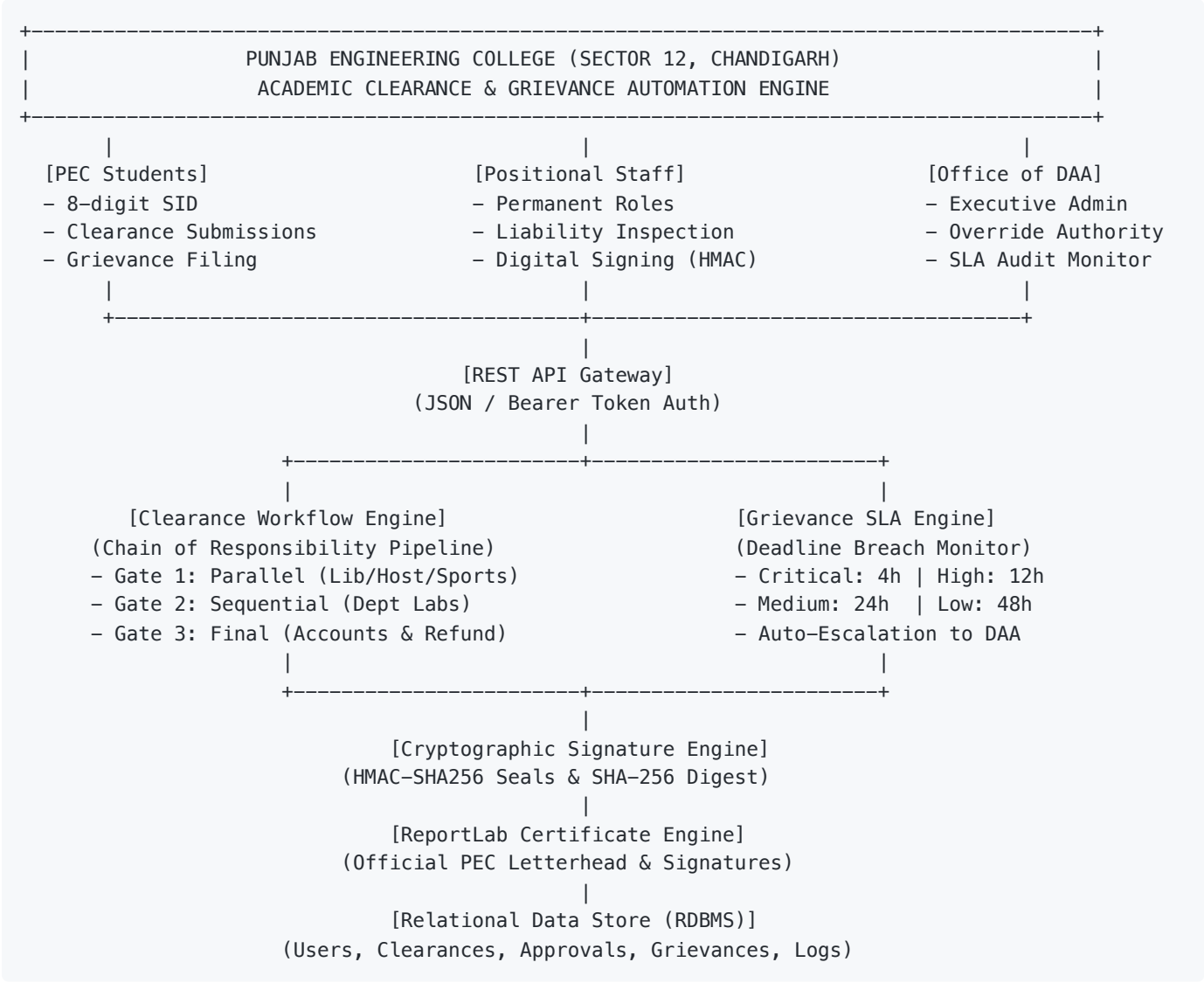
- **Section 2 (Overall Description):** Presents product perspective, context diagrams, major system functions, user personas, institutional constraints, and operational dependencies.
- **Section 3 (Specific Requirements):** Formally specifies external interface requirements, functional requirements organized into six discrete modules (`FR-1.1` to `FR-6.3`), performance benchmarks, design constraints, software quality attributes, and architectural invariants.
- **Section 4 (Appendix):** Provides the formal IEEE Std 830-1998 Quality and Verification Matrix.

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2. Overall Description

2.1 Product Perspective

The Clearance Automation Engine serves as the core operational platform mediating between graduating students, academic departments, and the Dean Academic Affairs (DAA) administration:



2.2 Product Functions (High-Level Summary)

The platform delivers six major functional capabilities:

1. **Role-Based Authentication & Session Scoping:** Validates institutional credentials and issues scoped JWT bearer tokens.
2. **Clearance Request Gating:** Instantiates multi-gate approval chains enforcing parallel and sequential dependencies.
3. **Cryptographic Sign-Off & Verification:** Generates HMAC-SHA256 seals per department review and SHA-256 certificate digests.
4. **Administrative Overrides:** Empowers the DAA to resolve department impasses with mandatory immutable audit logging.
5. **Grievance Redressal & SLA Escalation:** Tracks student appeals with automated timer expiration and escalation alerting.
6. **Certificate Minting & Verification:** Emits print-ready ReportLab vector PDFs and exposes unauthenticated public verification.

2.3 User Classes & Characteristics

1. **PEC Student Persona:**

- Identified by an 8-digit PEC SID (`22103042`) and `@pec.edu.in` email.
- Authorized to initiate one active clearance request per academic milestone.
- Monitors live gate progress and departmental liabilities in INR (`₹`).
- May submit formal priority-ranked grievances.

2. **Positional Staff Persona:**
   - Permanent administrative titles (*Librarian*, *Chief Warden*, *Gymkhana President*, *HOD Labs*, *Accounts Officer*).
   - Verifies physical asset returns, records departmental dues in INR (`₹`), and generates cryptographic digital seals.
   - Evaluates and settles student grievances.
3. **Dean Academic Affairs (DAA / Admin Persona):**
   - Executive institutional authority with full read-write oversight across all departments.
   - Authorized to execute administrative overrides (`ADMIN_OVERRIDE`) to unblock disputed approvals.
   - Monitors SLA compliance analytics and department bottlenecks.
4. **Public Verifier Persona:**
   - Third-party entities (employers, foreign universities, background check agencies).
   - Requires zero authentication credentials to validate certificate integrity via SHA-256 digests.

2.4 General Constraints

1. **Institutional Data Rules:**
   - Student IDs must adhere strictly to the 8-digit alphanumeric PEC format: `YYBBBRRR` .
   - All financial dues, fees, and caution deposit refunds must be denominated in Indian Rupees (`₹` / INR).
   - Staff credentials map to institutional positional titles rather than personal faculty names.
2. **Pipeline Ordering Invariants:**
   - Gate 2 (Department Labs) cannot unlock until all Gate 1 departments are `APPROVED` .
   - Gate 3 (Accounts) cannot unlock until Gates 1 and 2 are fully `APPROVED` .
   - Degree clearance certificates cannot be issued unless all 5 departmental gates reach `APPROVED` status.

2.5 Assumptions & Dependencies

1. **Institutional Network Availability:** The system operates over standard TCP/IP networking with HTTPS/TLS 1.3 connectivity.
2. **Server Time Synchronization:** System servers maintain Network Time Protocol (NTP) synchronization to ensure accurate SLA deadline computations.
3. **Client Capability:** Client browsers support ECMAScript 2022+ and modern HTML5 canvas/SVG rendering.

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3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

- **Student Dashboard:** Responsive Single Page Application (SPA) displaying the 3-gate visual pipeline, itemized liabilities in INR (`₹`), grievance dispute forms, and clearance certificate downloads.
- **Department Staff Console:** Department-filtered review queue enabling liability verification, dues input in INR (`₹`), resolution notes, and one-click cryptographic HMAC signing.
- **Dean Academic Affairs (DAA) Portal:** Executive analytics dashboard displaying institutional bottleneck metrics, overdue SLA breach alerts, student rosters, and administrative override controls.
- **Public Certificate Verification Interface:** Tokenless, unauthenticated web form accepting a certificate ID or SHA-256 hash and rendering instant verification credentials.

3.1.2 Hardware Interfaces

- **Server Tier:** Architecture-neutral x86-64 / ARM64 execution environment (minimum 1 vCPU, 512 MB RAM).
- **Client Tier:** Standard desktop workstation or mobile device browser with display resolution $\geq 375 \times 667$ px.

3.1.3 Software Interfaces

- **Operating System:** Platform independent (macOS, Linux / POSIX, Windows).
- **Application Server Gateway:** ASGI-compliant HTTP server (Uvicorn / FastAPI).
- **Database Engine Interface:** SQL-standard Relational Database Management System (SQLite3 embedded / PostgreSQL 14+) interfaced via SQLAlchemy Object-Relational Mapping (ORM).
- **Cryptographic Subsystem:** FIPS-compliant SHA-256 and HMAC-SHA256 hashing primitives.
- **Document Rendering Subsystem:** ReportLab vector graphics engine for high-resolution PDF certificate generation.

3.1.4 Communications Interfaces

- **Network Protocol:** HTTP/1.1 and HTTP/2 over TLS 1.3 (HTTPS).
- **Data Exchange Format:** JavaScript Object Notation (JSON, RFC 8259) with UTF-8 character encoding.
- **Authorization Header:** Standard HTTP `Authorization: Bearer <JWT_TOKEN>` for stateless session validation.

3.2 Functional Requirements (System Features)

3.2.1 Module 1: Authentication & Identity Management

- **FR-1.1 [Institutional Credential Verification]:** The system shall authenticate users using institutional credentials. Passwords shall be hashed using salted one-way hashing algorithms (Bcrypt).
- **FR-1.2 [Role & Scope Enforcement]:** The system shall enforce role-based access control (RBAC) across three principal scopes: `STUDENT`, `STAFF`, and `ADMIN`.
- **FR-1.3 [Positional Designation Binding]:** Staff profiles shall be bound to institutional department codes (`LIBRARY`, `HOSTEL`, `SPORTS`, `LABS`, `ACCOUNTS`) and permanent positional titles rather than personal identities.

3.2.2 Module 2: Clearance Request Lifecycle

- **FR-2.1 [Single Active Clearance Constraint]:** A student may possess at most one active clearance request in `PENDING` status at any given time.
- **FR-2.2 [Milestone Justification Recording]:** The clearance request must record the academic milestone:
  - B.Tech Final Degree Graduation
  - Mandatory 6-Month Industrial Training / Internship (Semester 7/8)
  - M.Tech / Ph.D. Thesis Submission
  - Program Withdrawal / Caution Money Settlement
- **FR-2.3 [Automated Approval Gate Spawning]:** Upon creation of a clearance request, the system shall instantiate exactly five departmental clearance records:
  - `LIBRARY` (Gate 1, initialized to `PENDING`)
  - `HOSTEL` (Gate 1, initialized to `PENDING`)
  - `SPORTS` (Gate 1, initialized to `PENDING`)
  - `LABS` (Gate 2, initialized to `WAITING_PREREQUISITE`)
  - `ACCOUNTS` (Gate 3, initialized to `WAITING_PREREQUISITE`)

3.2.3 Module 3: Chain of Responsibility (CoR) Approval Gating

- **FR-3.1 [Gate 1 Concurrent Evaluation]:** The Library, Hostel, and Sports Complex handlers shall evaluate and update clearance statuses concurrently and independently.
- **FR-3.2 [Gate 2 Prerequisite Enforcement]:** The Departmental Labs handler shall reject approval attempts if any Gate 1 department status is not `APPROVED`, unless an explicit `ADMIN_OVERRIDE` has been executed by the DAA.
- **FR-3.3 [Gate 3 Financial & Caution Money Settlement]:** The Accounts handler shall reject approval attempts if Gate 1 or Gate 2 is incomplete. Upon approval, Accounts reconciles institutional fees and authorizes the release of the student's Caution Money deposit.
- **FR-3.4 [Digital Signature Generation]:** Upon approving any gate, the handler shall compute a keyed digital signature seal: $\text{Seal} = \text{HMAC-SHA256}\Big(K_{\text{secret}}, ;\text{staff_id} \parallel \text{dept_code} \parallel \text{request_id} \parallel \text{timestamp}\Big)$
- **FR-3.5 [Terminal Status Propagation]:** When Gate 3 reaches `APPROVED`, the master clearance request status shall transition to `APPROVED`, and the system shall automatically mint a unique Certificate ID (`PEC-CLR-YYYY-XXXXX`) and an immutable SHA-256 certificate hash.

3.2.4 Module 4: Administrative Authority & Concurrency Control

- **FR-4.1 [Executive Override Authority]:** The Dean Academic Affairs (Admin) shall have authority to override any departmental rejection, transitioning the gate to `APPROVED` with an audit flag `ADMIN_OVERRIDE`.
- **FR-4.2 [Atomic Gate Recalculation]:** All prerequisite checks and gate unlocking transitions shall execute within serialized database transactions to prevent race conditions during concurrent staff reviews.
- **FR-4.3 [Audit Trail Immutability]:** Every approval, rejection, due modification, grievance resolution, and override shall append an immutable record to the institutional audit log containing timestamp, actor ID, action type, and JSON metadata.

3.2.5 Module 5: Grievance Arbitration & Dynamic SLA Escalation

- **FR-5.1 [Grievance Filing]:** A student may lodge a formal grievance against any department reporting outstanding dues, specifying subject, description, priority, and disputed amount in INR (`₹`).

- **FR-5.2 [Dynamic SLA Computation]:** The system shall compute an exact resolution deadline upon grievance registration based on priority: $$\text{Deadline} = \text{Timestamp}*{|text{submission}} + |Delta t*{\text{priority}}$$
  - `CRITICAL` : $\Delta t = 4$ hours
  - `HIGH` : $\Delta t = 12$ hours
  - `MEDIUM` : $\Delta t = 24$ hours
  - `LOW` : $\Delta t = 48$ hours
- **FR-5.3 [Automated DAA Escalation Engine]:** The system shall evaluate pending grievances against the current timestamp. Any grievance where $\text{Current Time} > \text{Deadline}$ with status `OPEN` shall transition to `ESCALATED`, setting `is_escalated = True` and notifying the Dean Academic Affairs dashboard.
- **FR-5.4 [Grievance Resolution]:** The concerned department in-charge or DAA may resolve a grievance with resolution notes, optionally updating or waiving the associated dues.

3.2.6 Module 6: Certificate Generation & Public Verification

- **FR-6.1 [Cryptographic Hash Calculation]:** The certificate hash shall be computed as: \text{Hash}_{\text{cert}} = \text{SHA-256}\Big(\text{cert_id} \parallel \text{student_sid} \parallel \text{academic_year} \parallel \text{completion_timestamp}\Big)
- **FR-6.2 [Official PDF Certificate Rendering]:** The system shall dynamically generate a print-ready vector PDF document bearing:
  - Official Punjab Engineering College header, crest, and Sector 12 address.
  - Student details (Full Name, 8-digit SID, Academic Program, Branch).
  - Departmental clearance breakdown with timestamps and positional titles.
  - Caution Money refund confirmation in INR (`₹`).
  - Cryptographic verification SHA-256 hash.
  - Signature blocks for Dean Academic Affairs (DAA) and Dean Student Affairs (DSA).
- **FR-6.3 [Public Hash Verification]:** The system shall provide a public REST endpoint `/api/v1/certificate/verify` accepting `certificate_id` or `certificate_hash`. It shall return verification status, student name, SID, issuance date, and validity confirmation without requiring user authentication.

3.3 Performance Requirements

- **NFR-1.1 [API Latency]:** Under normal operating load (100 concurrent requests), 95% of read queries shall respond within ≤ 150 ms, and 95% of state modification transactions shall complete within ≤ 300 ms.
- **NFR-1.2 [Certificate Rendering Throughput]:** Dynamic vector PDF generation shall complete in ≤ 1.5 seconds.
- **NFR-1.3 [SLA Evaluation Execution]:** The automated SLA escalation scan shall complete evaluation across all active tickets in ≤ 500 ms.

3.4 Design Constraints

- **NFR-2.1 [Database Agnosticism]:** Persistence logic must use SQLAlchemy ORM abstractions without raw dialect-specific SQL, allowing transparent switching between SQLite3 and PostgreSQL.
- **NFR-2.2 [Zero External Authentication Dependencies]:** The system shall maintain self-contained JWT authentication without external SSO dependencies.

- **NFR-2.3 [Input Sanitization & Injection Prevention]:** All database operations must be executed through parameterized ORM statements to eliminate SQL injection vulnerabilities.

3.5 Software System Attributes

3.5.1 Security Requirements

- **NFR-3.1 [Token-Based Authentication]:** API communication shall be authenticated via cryptographically signed JWT bearer tokens with expiration limits.
- **NFR-3.2 [Cryptographic Signatures]:** Departmental seals must use HMAC-SHA256 with a private institutional server key. Any tampering with stored approval records shall invalidate the digital signature.
- **NFR-3.3 [Role-Based Access Isolation]:** Students cannot query or mutate records belonging to other students; staff members can only review approvals matching their assigned department.

3.5.2 Concurrency & Idempotency Requirements

- **NFR-4.1 [Concurrent Approval Consistency]:** Concurrent approvals by Gate 1 departments (Hostel, Library, Sports) shall execute idempotently without deadlocks or lost updates.
- **NFR-4.2 [Client-Side Synchronization]:** Client dashboards shall synchronize with the server state within an interval of $\leq 10\ \text{seconds}$ to reflect cross-user updates.

3.5.3 Reliability & Availability

- **NFR-5.1 [Fault Isolation]:** Failure in one departmental workflow (e.g., temporary unavailability of Sports staff) shall not block independent operations in other Gate 1 departments.
- **NFR-5.2 [Transactional Integrity]:** Clearance request status updates and gate prerequisite unlocks shall be atomic (all-or-nothing transactions).

3.5.4 Maintainability & Extensibility

- **NFR-6.1 [Design Pattern Enforcement]:** The approval pipeline must adhere to the Chain of Responsibility pattern, allowing new approval gates to be inserted by extending `DepartmentHandler`.
- **NFR-6.2 [Static Code Quality Standards]:** The backend codebase shall maintain 0 violations under PEP 8 / Flake8 static analysis.

3.6 Architectural & System Invariants (Frozen Boundaries)

The following architectural invariants are **strictly frozen** and shall remain unchanged across any code refactoring or feature additions:

1. **Three-Gate Structural Topology:** The 3-gate progression ($\text{Gate 1 [Parallel]} \rightarrow \text{Gate 2 [Labs]} \rightarrow \text{Gate 3 [Accounts]}$) is invariant.
2. **Positional Staff Designations:** System roles shall remain role-based and positional (DAA, Chief Warden, Librarian, Gymkhana President, HOD Labs, Accounts Officer).
3. **Monetary Unit of Account:** The system's base currency is Indian Rupees (`₹` / INR).
4. **Institutional Identifier:** Student identification is bound to the 8-digit PEC SID format (`YYBBBRRR`).

5. **Cryptographic Validation Contract:** Certificate verification relies on SHA-256 digest comparison against immutable stored approval records.

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4. Appendix: IEEE Std 830-1998 Quality & Verification Matrix

This specification conforms strictly to the **IEEE Std 830-1998** standard quality requirements for well-formed software specifications:

IEEE 830 Quality Attribute	Specification Fulfillment	Section Reference	Verification Method
1. Correctness	Every stated requirement corresponds directly to institutional regulations of Punjab Engineering College (Sector 12, Chandigarh).	Sections 1.2, 3.2	Stakeholder Review
2. Unambiguity	Every functional requirement possesses an explicit, unique identifier (`FR-X.Y`) with quantifiable inputs, outputs, and mathematical formulas.	Section 3.2	Traceability Matrix
3. Completeness	Covers all institutional clearance departments, fee reconciliation, SLA grievance tracking, and cryptographic certificate minting.	Sections 2.1, 3.2	Requirement Coverage
4. Consistency	Gate progression constraints, actor permissions, and state transitions are mathematically non-contradictory across all modules.	Sections 3.2, 3.6	Formal Model / UML State
5. Ranked for Importance / Stability	Core 3-gate pipeline and cryptographic seals are designated as **Essential & Frozen Invariants**; analytics and export tools are supportive.	Section 3.6	Priority Classification
6. Verifiability	Every functional and non-functional requirement is objectively testable via automated unit tests, static analysis, or end-to-end API calls.	Sections 3.2, 3.3	Automated Test Suite (11/11 Passed)
7. Modifiability	Organized hierarchically into decoupled modules, allowing implementation adjustments without altering requirements contracts.	Sections 3.1 - 3.2	Modular Structure
8. Traceability	Forward and backward traceability maintained from user personas $\rightarrow$ functional requirements $\rightarrow$ DFD processes $\rightarrow$ UML classes.	Sections 2.3, 3.2	Traceability Matrix